

Azaria Reed

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WORK EXPERIENCE

Systems Engineer I

SafeHug LLC

2022 – Present

Huntsville, AL

- Owned verification and validation of a production web system supporting a patented medical safety device, ensuring system behavior consistently aligned with real-world safety intent
- Validated end-to-end system functionality, performance characteristics, and operational reliability, translating stakeholder intent into verifiable technical outcomes
- Led change management and continuous monitoring, proactively identifying risks, anomalies, and emerging issues to preserve system stability, security, and operability
- Maintained long-term system reliability through structured validation checks, controlled updates, and ongoing verification of deployed behavior
- Served as the primary technical point of contact between engineering implementation and inventor intent, ensuring traceability between requirements and system behavior

Software Engineering Intern

Tektonux LLC

May 2025 – Aug. 2025

Huntsville, AL

- Implemented user interfaces in TypeScript, CSS, and HTML by communicating directly with the design team to ensure accuracy at each stage of the development process
- Supported the design and execution of automated tests with Playwright to enable the validation of software behavior relative to visual designs
- Tested a terrain server using GeoTIFF DEMs and Cesium to verify that its functionality was replicable on new machines
- Drafted a user manual for a Cyber Dashboard to communicate system behavior clearly to end users
- Composed documentation for unit tests in C#, end-to-end tests, and Storybook to allow new developers to locate, understand, and employ tests efficiently
- Supplemented company onboarding documentation to support consistency and reliability in development practices
- Adapted to a professional team environment by incorporating feedback and contributing to cross-functional validation efforts

Test Engineer

Independent Small Business Client (E-Commerce)

Dec. 2022

Dallas, TX

- Validated and released a customer-facing production web system under strict time constraints, enabling immediate business deployment and revenue impact
- Tested system behavior, usability, and visual correctness to ensure production readiness prior to public launch
- Collaborated directly with the business owner to confirm requirements, rapidly resolve issues, and verify that delivered behavior matched business expectations

EDUCATION

University of Alabama in Huntsville

Bachelor of Science in Computer Science

Huntsville, AL

- Summa Cum Laude
- Minor: Mathematical Sciences
- Concentration: Artificial Intelligence and Data Science

Calhoun Community College

Associate of Science in Computer Science

Decatur, AL

- Summa Cum Laude | Top Ten Student Award
- Short-Term Certificate in Programming Essentials
- Short-Term Certificate in Computer Science Essentials

PROJECTS

[Machine Learning Projects](#)

- Designed and executed validation experiments for regression and classification models using accuracy, precision, and error metrics
- Analyzed model behavior and failure modes through diagnostic plots and comparative performance analysis
- Authored formal technical documentation detailing methodology, assumptions, results, and limitations

[UX Project](#)

- Conducted structured usability validation with real participants to assess system effectiveness and clarity
- Synthesized qualitative and quantitative findings into actionable, engineering-focused insights

CERTIFICATIONS, SKILLS & TECHNOLOGIES

- Certifications:** Certified Microsoft Office Specialist
- Core Engineering Skills:** Verification & Validation (V&V), Requirements Validation, Experimental Design, Metrics & Evaluation, Technical Documentation, Test Documentation
- Programming & Scripting:** Python, C++, C#, Java, TypeScript, HTML, CSS
- Technologies:** Playwright, GeoTIFF DEMs, Cesium